Chpater3 Electricity & Magnetism

Lesson1 Electrostatics & Coulomb's Law

1. Explain from an atomic viewpoint why charge is usually transferred by electrons.
2. A person is placed in a large, hollow metallic sphere that is insulated from ground. If a large charge is placed on the sphere, will the person be harmed upon touching the inside of the sphere?
3. Why must hospital personnel wear special conducting shoes while working around oxygen in an operating room? What might happen if the personnel wore shoes with rubber soles?
4. **(a)** Would life be different if the electron were positively charged and the proton were negatively charged? **(b)** Does the choice of signs have any bearing on physical and chemical interactions? Explain your answers.
5. If a suspended object A is attracted to a charged object B, can we conclude that A is charged? Explain.
6. Explain how a positively charged object can be used to leave another metallic object with a net negative charge. Discuss the motion of charges during the process.
7. Consider point A in **Figure CQ15.8** located an arbitrary distance from two-point charges in otherwise empty space. **(a)** Is it possible for an electric field to exist at point A in empty space? **(b)** Does charge exist at this point? **(c)** Does a force exist at this point?

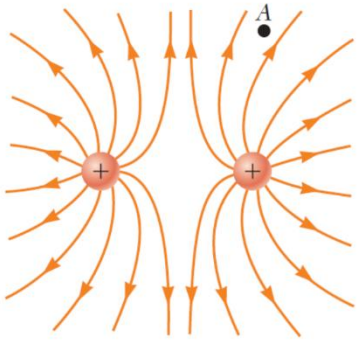


Figure CQ15.8

8. In fair weather, there is an electric field at the surface of the Earth, pointing down into the ground. What is the sign of the electric charge on the ground in this situation?
9. A charged comb often attracts small bits of dry paper that then fly away when they touch the comb. Explain why that occurs.
10. Why should a ground wire be connected to the metal support rod for a television antenna?
11. There are great similarities between electric and gravitational fields. A room can be electrically shielded so that there are no electric fields in the room by surrounding it with a conductor. Can a room be gravitationally shielded? Explain.
12. A spherical surface surrounds a point charge q. Describe what happens to the total flux through the surface if **(a)** the charge is tripled, **(b)** the volume of the sphere is doubled, **(c)** the surface is changed to a cube, **(d)** the charge is moved to another location inside the surface, and **(e)** the charge is moved outside the surface.
13. A glass object receives a positive charge of $13\ \text{nC}$ by rubbing it with a silk cloth. In the rubbing process, have protons been added to the object or have electrons been removed from it?
14. Two metal balls A and B of negligible radius are floating at rest on Space Station Freedom between two metal bulkheads, connected by a taut nonconducting thread of length $2.00\ \text{m}$. Ball A carries charge q, and

ball B carries charge 2q. Each ball is 1.00 m away from a bulkhead. **(a)** If the tension in the string is 2.50 N, what is the magnitude of q? **(b)** What happens to the system as time passes? Explain.

15. A 7.50-nC charge is located 1.80 m from a 4.20-nC charge. **(a)** Find the magnitude of the electrostatic force that one particle exerts on the other. **(b)** Is the force attractive or repulsive?

16. A charged particle A exerts a force of 2.62 N to the right on charged particle B when the particles are 13.7 mm apart. Particle B moves straight away from A to make the distance between them 17.7 mm. What vector force does particle B then exert on A?

17. A small sphere of mass $m = 7.50$ g and charge $q_1 = 32.0$ nC is attached to the end of a string and hangs vertically as in **Figure P15.4**. A second charge of equal mass and charge $q_2 = 258.0$ nC is located below the first charge a distance $d = 2.00$ cm below the first charge as in **Figure P15.4**. **(a)** Find the tension in the string. **(b)** If the string can withstand a maximum tension of 0.180 N, what is the smallest value d can have before the string breaks?

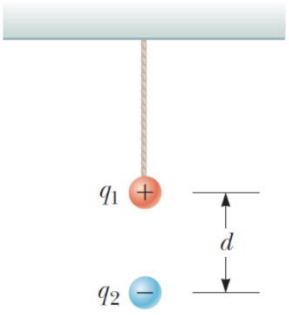


Figure P15.4

18. The nucleus of ^{8}Be, which consists of 4 protons and 4 neutrons, is very unstable and spontaneously breaks into two alpha particles (helium nuclei, each consisting of 2 protons and 2 neutrons). **(a)** What is the force between the two alpha particles when they are 5.00×10^{-15} m apart, and **(b)** what is the initial magnitude of the acceleration of the alpha particles due to this force? Note that the mass of an alpha particle is 4.002 6 u.

19. A small sphere of charge 0.800 μC hangs from the end of a spring as in **Figure P15.7a**. When another small sphere of charge 20.600 μC is held beneath the first sphere as in **Figure P15.7b**, the spring stretches by $d = 3.50$ cm from its original length and reaches a new equilibrium position with a separation between the charges of $r = 5.00$ cm. What is the force constant of the spring?

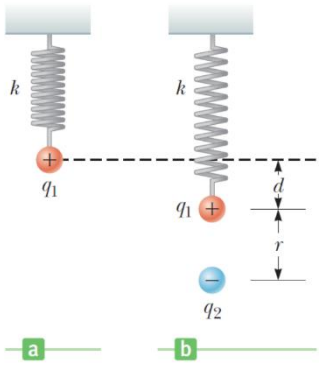


Figure P15.7

20. Four point charges are at the corners of a square of side a as shown in **Figure P15.8**. Determine the magnitude and direction of the resultant electric force on q, with k_e , q, and a left in symbolic form.

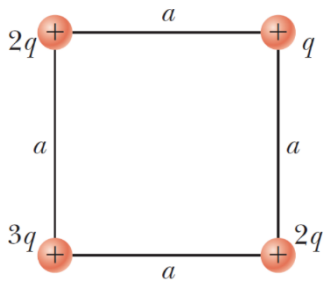


Figure P15.8

21. Two small identical conducting spheres are placed with their centers $0.30\ \text{m}$ apart. One is given a charge of 12×10^{-9} C, the other a charge of -18×10^{-9} C. **(a)** Find the electrostatic force exerted on one sphere by the other. **(b)** The spheres are connected by a conducting wire. Find the electrostatic force between the two after equilibrium is reached, where both spheres have the same charge.

22. Calculate the magnitude and direction of the Coulomb force on each of the three charges shown in **Figure P15.10.**

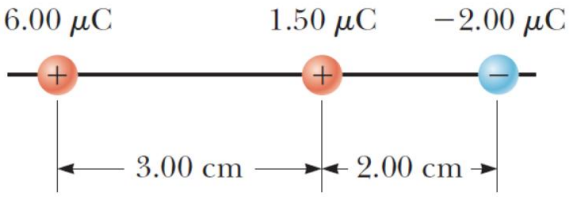


Figure P15.10 Problems 10 and 18.

23. Three charges are arranged as shown in **Figure P15.11**. Find the magnitude and direction of the electrostatic force on the charge at the origin.

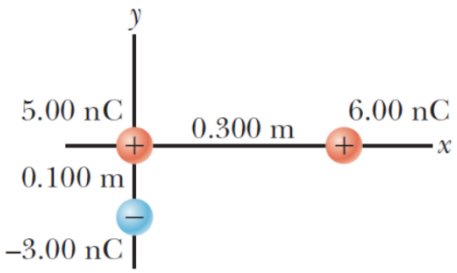


Figure P15.11

24. A positive charge $q_1 = 2.70\ \mu\text{C}$ on a frictionless horizontal surface is attached to a spring of force constant k as in **Figure P15.12**. When a charge of $q_2 = 28.60\ \mu\text{C}$ is placed $9.50\ \text{cm}$ away from the positive charge, the spring stretches by $5.00\ \text{mm}$, reducing the distance between charges to $d = 9.00\ \text{cm}$. Find the value of k.

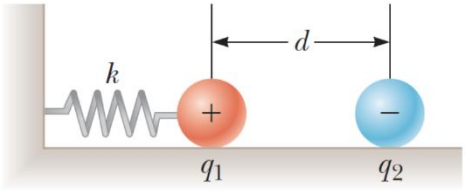


Figure P15.12

25. Three-point charges are located at the corners of an equilateral triangle as in **Figure P15.13**. Find the magnitude and direction of the net electric force on the 2.00 μC charge.

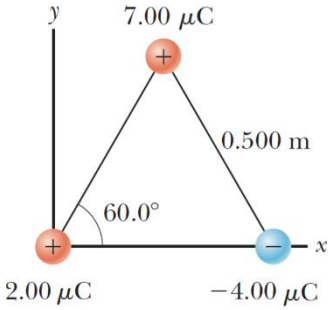


Figure P15.13 Problems 13 and 24.

26. A small object of mass 3.80 g and charge 218 μC is suspended motionless above the ground when immersed in a uniform electric field perpendicular to the ground. What is the magnitude and direction of the electric field?

27. An electric field of magnitude 5.25×10^5 N/C points due south at a certain location. Find the magnitude and direction of the force on a 26.00 μC charge at this location.

28. An electron is accelerated by a constant electric field of magnitude 300 N/C. **(a)** Find the acceleration of the electron. **(b)** Use the equations of motion with constant acceleration to find the electron's speed after 1.00×10^{-8} s, assuming it starts from rest.

29. In **Figure P15.27** determine the point (other than infinity) at which the total electric field is zero.

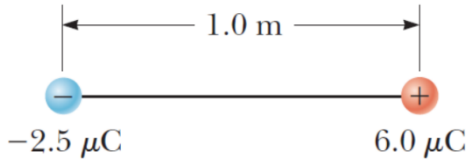


Figure P15.27

30. Three charges are at the corners of an equilateral triangle, as shown in **Figure P15.28**. Calculate the electric field at a point midway between the two charges on the x-axis.

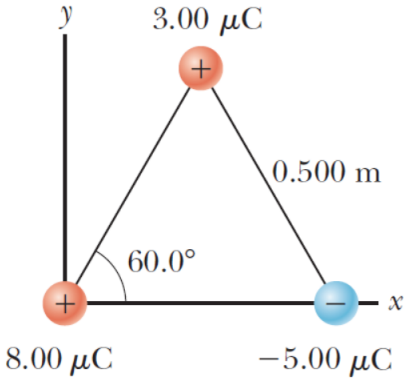


Figure P15.28